v4 Post-processed Results — P1 / P3 / P1+P3

2026-07-27 · Companion to `RESULTS_BD_V4.pdf`, which reports the same runs **without any post-processing**. Nothing here is a new experiment: every number is the post-processing stage of the runs already described there (same v4 EOS-fixed backbones, same discriminators/classifiers, same 1,000 reserved except_0 pairs, seed 7, $\gamma = 1$). Split into its own file so the RAW report stays clean.

1. What P1 and P3 do, and how to read these tables

- **P1 (edge splice)**: every illegal transition (u, v) in a generated path is replaced by a shortest detour u → v on the exceptional graph $A_{except0}$. By construction **validity becomes 1.000**; the endpoint is untouched, so **arrival is unchanged from RAW**.
- **P3 (endpoint patch)**: a shortest path from the generated endpoint to the true destination is appended. By construction **arrival becomes 1.000**; illegal edges elsewhere remain, so **validity is unchanged from RAW**.
- **P1+P3**: both, in that order — validity and arrival are both 1.000 for every method and every block.

> **Consequence for reading:** under P1+P3 the two headline RAW metrics are pinned at 1.000 and carry no information. The columns that still discriminate are **EM** (exact-match with a true shortest path), **PC** (path-correctness rank score) and **patch** — the mean number of hops the repair had to insert, i.e. how much of the final path was produced by Dijkstra rather than by the model. A method that generates badly can still reach EM parity by being patched heavily; the patch column is what keeps that honest.

All P1/P3/P1+P3 numbers were recorded by the evaluation runs themselves (`three_way_postproc.py` emits raw/P1/P3/P1P3, `eval_dcbg.py` emits raw/P1P3) and collected with `collect_postproc.py` — no recomputation. Raw artifact: `sets_res/postproc_v4.json` (336 guided rows + 56 D-CBG rows, none missing).

2. fam 0.05 — P1+P3 (valid = arrival = 1.000 throughout)

2.1 Seen (disc trained on except_0)

blk	base EM	base PC	base patch	IW EM	IW PC	IW patch	adj+IW EM	adj+IW PC	adj+IW patch
1	0.300	0.452	5.56	0.307	0.459	5.33	**0.376**	0.432	8.59
2	0.267	0.421	6.70	0.325	0.472	5.69	**0.382**	**0.513**	5.17
4	0.272	0.436	5.64	0.312	0.467	5.32	**0.342**	0.487	5.00
8	0.253	0.418	6.13	0.300	0.450	5.75	**0.323**	0.470	5.47
16	0.231	0.394	6.71	0.243	0.406	6.35	**0.266**	0.429	5.62
32	0.219	0.380	7.00	0.243	0.404	6.64	**0.259**	0.418	6.26
64	0.194	0.353	8.83	0.211	0.372	7.81	**0.230**	0.394	7.40

2.2 Unseen (disc trained on except 1–99, zero-shot; base identical to §2.1)

blk	base EM	base PC	base patch	IW EM	IW PC	IW patch	adj+IW EM	adj+IW PC	adj+IW patch
1	0.300	0.452	5.56	0.305	0.457	5.33	**0.364**	0.419	8.80
2	0.267	0.421	6.70	0.325	0.474	5.39	**0.367**	**0.502**	4.95
4	0.272	0.436	5.64	0.311	0.467	5.36	**0.345**	0.488	5.01
8	0.253	0.418	6.13	0.303	0.451	5.78	**0.323**	0.469	5.44
16	0.231	0.394	6.71	0.251	0.412	6.32	**0.272**	0.434	5.60
32	0.219	0.380	7.00	0.243	0.405	6.61	**0.258**	0.418	6.23
64	0.194	0.353	8.83	0.213	0.373	7.80	**0.233**	0.395	7.46

- **The RAW ordering survives post-processing**: adj+IW > IW > base in EM at every block, in both regimes. Guidance is not something P1+P3 can substitute for — repairing a worse path yields a worse final path.
- **adj+IW also patches least** at blk2–64 (e.g. blk64 7.40 vs base 8.83 hops), so its EM lead is not bought with extra Dijkstra. **blk1 is the exception**: adj+IW needs 8.59 hops vs base 5.56, because its RAW arrival is only 0.567 (§4.1 of the RAW report) and P3 has to walk the rest — its PC (0.432) actually falls below IW's (0.459) for that reason, even though EM is highest.
- **Post-processing compresses the block-size spread**: RAW EM ranges 0.129–0.271 across blocks (base), P1+P3 EM 0.194–0.300 — Dijkstra repairs absorb part of the large-block degradation, at a visible patch cost (8.83 hops at blk64 vs 5.56 at blk1).

- **Seen/unseen remains within ≤0.008 EM** everywhere, so zero-shot transfer holds after repair too.

3. fam 0.05 — P1 alone and P3 alone

Isolating the two repairs (EM, seen). **P1** forces validity to 1.000 and leaves arrival at its RAW value; **P3** forces arrival to 1.000 and leaves validity at its RAW value.

blk	base P1	base P3	IW P1	IW P3	adj+IW P1	adj+IW P3
1	0.300	0.271	0.307	0.283	**0.377**	0.373
2	0.272	0.229	0.334	0.297	**0.402**	0.369
4	0.278	0.238	0.323	0.285	**0.361**	0.315
8	0.264	0.196	0.313	0.258	**0.336**	0.283
16	0.237	0.177	0.256	0.203	**0.288**	0.216
32	0.220	0.156	0.253	0.194	**0.276**	0.204
64	0.202	0.125	0.225	0.152	**0.246**	0.168

- **P1 does nearly all the work**: at blk4 seen, adj+IW EM is 0.361 with P1 alone vs 0.342 with P1+P3 — adding P3 *lowers* EM, because forcing arrival appends hops that break exact length matching. The same holds for every method and block.
- **P3 alone is the weakest repair** (blk64 base 0.125 vs RAW 0.129): patching the endpoint without fixing illegal edges leaves the path invalid, and EM requires validity.
- So the useful configurations are **P1 alone** (best EM) or **P1+P3** (EM slightly lower but arrival guaranteed) — P3 alone is dominated.

4. fam 0.1 — P1+P3

Edge-remove ratio 0.1, i.e. twice the removed edges. Same layout; EM / PC / patch, seen and unseen.

blk	base EM	IW EM	adj+IW EM	base PC	IW PC	adj+IW PC	base patch	IW patch	adj+IW patch
1	**0.147**	0.142	0.143	**0.262**	0.260	0.178	14.65	13.86	17.03
2	0.135	0.153	**0.188**	0.251	0.271	**0.298**	15.55	14.53	13.21
4	0.119	0.145	**0.166**	0.240	0.263	**0.283**	14.52	14.46	12.94
8	0.118	0.139	**0.150**	0.235	0.256	**0.273**	15.81	15.44	13.88
16	0.103	0.119	**0.138**	0.224	0.235	**0.260**	16.32	15.22	13.74
32	**0.111**	0.104	0.109	0.223	0.221	**0.236**	16.90	16.40	14.82
64	0.089	0.108	**0.132**	0.210	0.227	**0.255**	17.49	17.20	15.23

Unseen mirrors seen to ≤0.004 EM at every block (blk1 0.147/0.141/0.144, blk64 0.089/0.105/0.133).

- **Repair now dominates the output**: patch is 13–17 hops against a real path length of ~23, i.e. more than half of the final path is Dijkstra's, not the model's. EM ≈ 0.09–0.19 despite valid = arrival = 1.000. At this difficulty the post-processed numbers say more about the repair than about the generator.
- **adj+IW still leads on PC at every block** and on EM everywhere except blk1 and blk32; the blk1 anomaly is the arrival collapse again (RAW 0.250), which forces a 17-hop patch and drags PC to 0.178 — the worst cell in the table despite a competitive EM.
- **Guidance matters more, not less, after repair** at large blocks (blk64 EM 0.089 → 0.132, +48% relative) — the opposite of the fam 0.05 picture, where post-processing narrowed the gaps.

5. D-CBG — P1+P3 (fam 0.05, γ = 1)

`eval_dcbg.py` emits raw and P1+P3 only (no P1/P3 in isolation). EM / PC / patch; the IW and adj+IW columns from §2.1 are repeated for reference.

blk	D-CBG exact EM (seen/unseen)	D-CBG 1st-order EM (seen/unseen)	IW EM (seen)	adj+IW EM (seen)	D-CBG exact PC / patch (seen)
1	0.307 / 0.302	0.301 / 0.300	0.307	**0.376**	0.460 / 5.19
2	0.285 / 0.282	0.269 / 0.268	0.325	**0.382**	0.438 / 6.27
4	0.282 / 0.279	0.276 / 0.274	0.312	**0.342**	0.443 / 5.52
8	0.256 / 0.260	0.253 / 0.252	0.300	**0.323**	0.422 / 6.03
16	0.245 / 0.241	0.232 / 0.232	0.243	**0.266**	0.407 / 6.41
32	0.235 / 0.224	0.219 / 0.219	0.243	**0.259**	0.400 / 6.47
64	0.200 / 0.200	0.195 / 0.194	0.211	**0.230**	0.362 / 8.06

- **The RAW conclusion is unchanged by post-processing**: adj+IW > IW ≥ D-CBG exact > D-CBG first-order ≈ base in EM at almost every block. (blk16/32 are the exception, where D-CBG exact edges out plain IW by 0.002 EM — inside run-to-run noise.)
- **First-order stays indistinguishable from base after repair too** (e.g. blk64 0.195 vs base 0.194; blk4 0.276 vs 0.272), confirming §5.3–§5.4 of the RAW report on the post-processed metrics as well.
- D-CBG's seen/unseen gap after repair is ≤0.011 EM, in line with everything else in this round.

6. Notes

- P1/P3 use `networkx.shortest_path` on the exceptional graph $G_{except0}$; when no path exists the segment is left as-is (rare, and it is why validity is 1.000 rather than merely near-1.000 in these runs — no such failure occurred).
- **patch** is the mean number of hops *added* by the repair, averaged over all 1,000 paths (0.00 for RAW by definition).
- Sources: `sets_log/v4_eval_job*.log` (guided, 336 rows), `sets_log/v4_dcbgeval_job*.log` (D-CBG, 56 rows), collected by `collect_postproc.py` into `sets_res/postproc_v4.json`.
- The RAW counterparts of every table here are in `RESULTS_BD_V4.pdf` §4 (fam 0.05), §5 (D-CBG) and §6 (fam 0.1); discriminator-level analysis is in `DISC_VALIDITY.pdf`.